AMAZON.COM RECOMMENDATIONS

ITEM-TO-ITEM COLLABORATIVE FILTERING

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OUTLINE

- INTRODUCTION
- DIFFERENT TYPES OF FILTERING SYSTEMS
- RECOMMENDATION ALGORITHM CHALLENGES
- USER AND ITEM BASED FILTERING
- TRADITIONAL COLLABRATIVE FILTERING
- CLUSTER MODELS
- SEARCH BASED METHODS
- ITEM-TO-ITEM COLLABRATIVE FILTERING
- SCALABILITY
- CONCLUSION

Recommendation algorithms are best known for their use on e-commerce websites where they use input about a customer's interests to generate a list of recommendation items.

Many applications use only the items that customers purchase and explicitly rate to represent their interests, but they can also use other attributes, including items viewed, demographic data, subject interests, and favorite artists.

35 % of amazon revenue is from recommendation engine

INTRODUCTION

1. Collaborative filtering.
2. Cluster models.
3. Search based methods.
4. Demographic based Recommender system.
5. Knowledge based Recommender system.
6. Hybrid filtering.

DIFFERENT TYPES OF FILTERING SYSTEMS

RECOMMENDATION ALGORITHM CHALLENGES

A LARGE RETAILER MIGHT HAVE HUGE AMOUNTS OF DATA, TENS OF MILLIONS OF CUSTOMERS AND MILLIONS OF DISTINCT CATALOG ITEMS.

MANY APPLICATIONS REQUIRE THE RESULTS SET TO BE RETURNED IN REAL TIME, IN MORE THAN HALF A SECOND WHILE STILL PRODUCING HIGH QUALITY RECOMMENDATIONS.

NEW CUSTOMERS TYPICALLY HAVE EXTREMELY LIMITED INFORMATION BASED ON ONLY A FEW PURCHASES OR PRODUCT RATINGS.

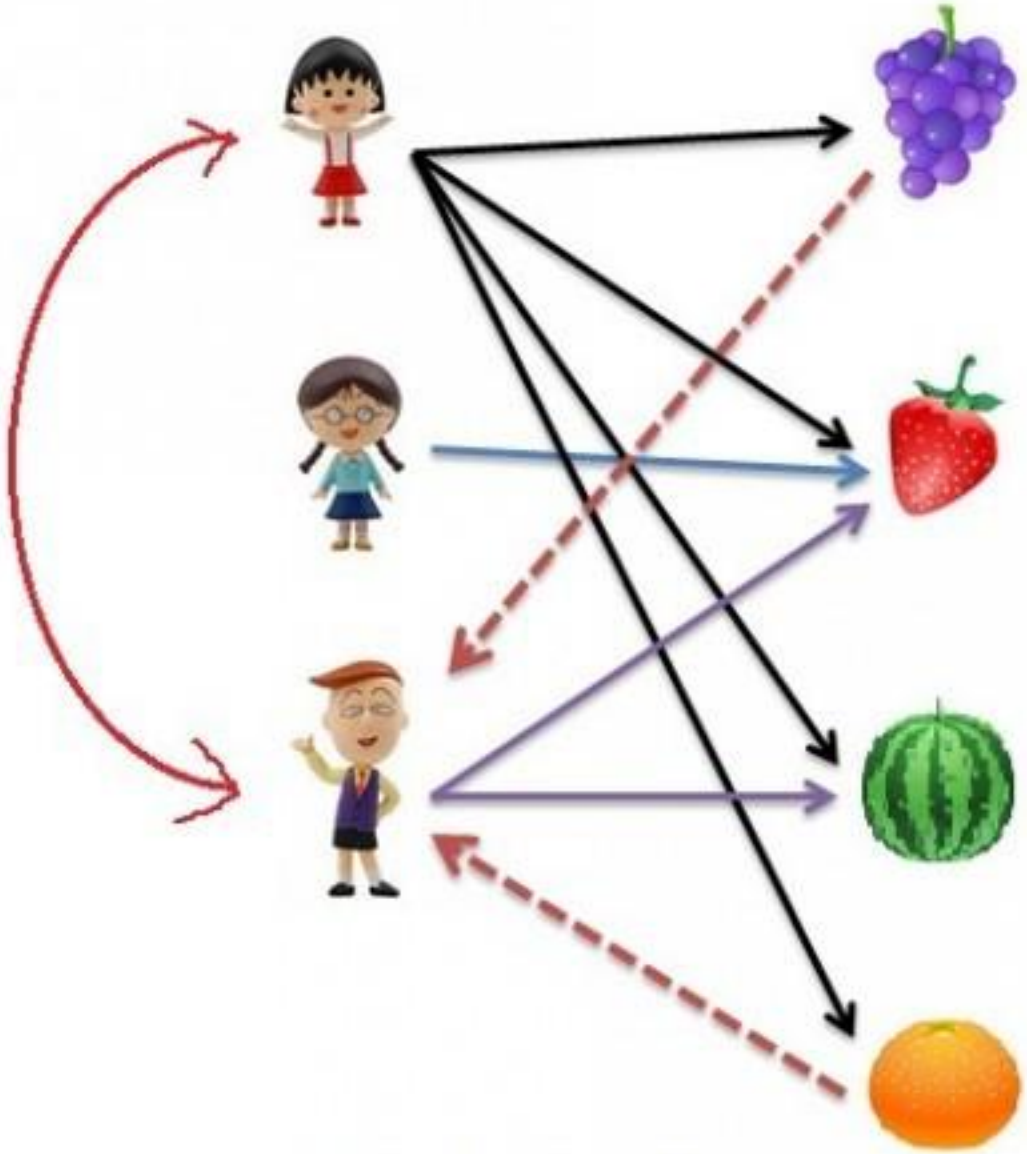
OLDER CUSTOMER CAN HAVE A GLUT OF INFORMATION BASED ON THOUSANDS OF PURCHASES AND RATINGS.

CUSTOMER DATA IS VOLATILE: EACH INTERACTION PROVIDES VALUABLE CUSTOMER DATA, AND THE ALGORITHM MUST RESPOND IMMEDIATELY TO THE NEW INFORMATION.

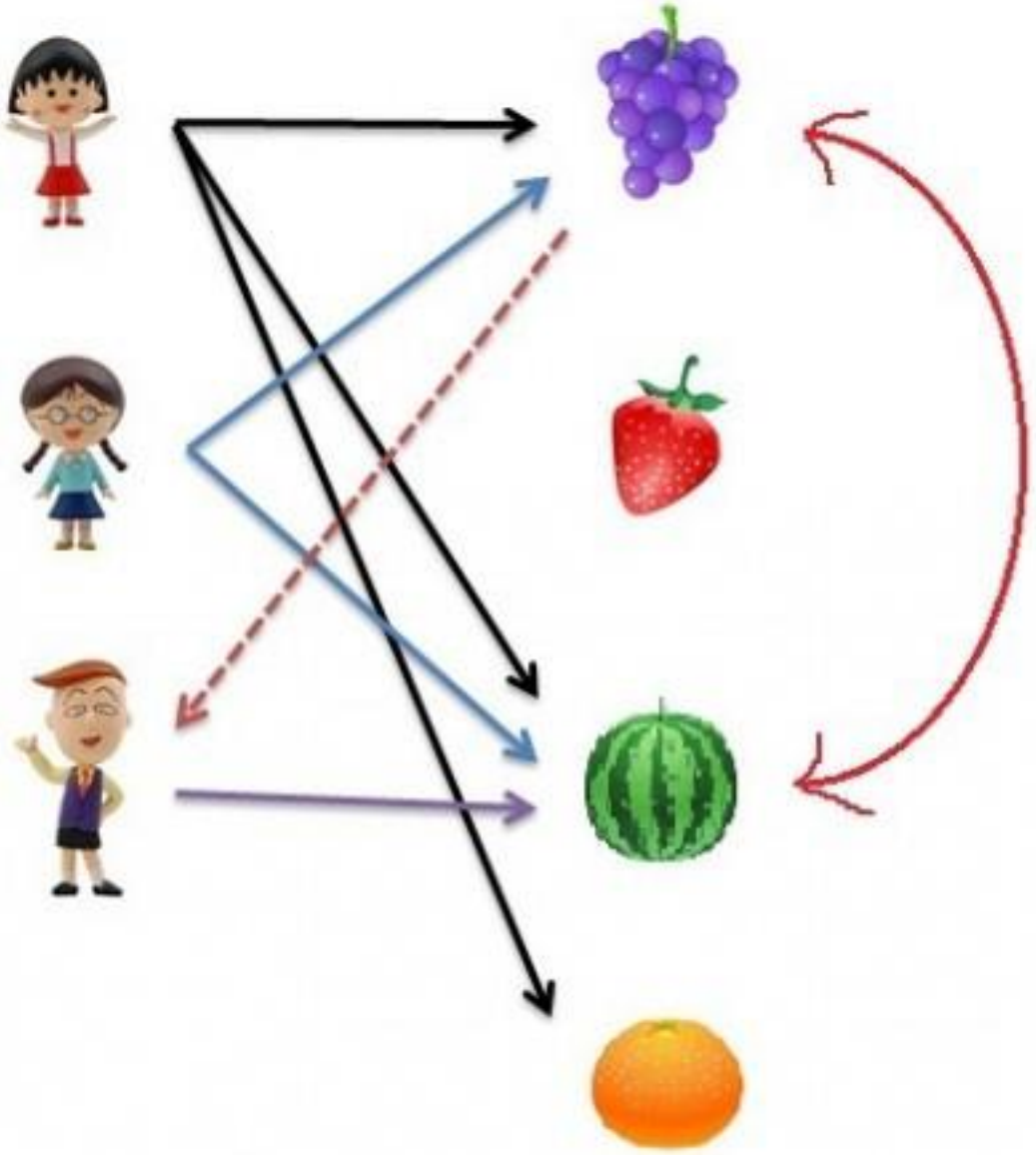
COLLABORATIVE FILTERING AND CLUSTER MODELS FOCUS ON FINDING SIMILAR SET OF CUSTOMERS WHOSE PURCHASED AND RATED ITEMS OVERLAP WITH THE USER'S PURCHASED AND RATED ITEMS.

SEARCH-BASED METHODS AND ITEM-TO-ITEM COLLABORATIVE FILTERING FOCUS ON FINDING SIMILAR ITEMS, NOT SIMILAR CUSTOMERS.

USER BASED AND ITEM BASED FILTERING



User-based filtering



Item-based filtering

TRADITIONAL COLLABORATIVE FILTERING

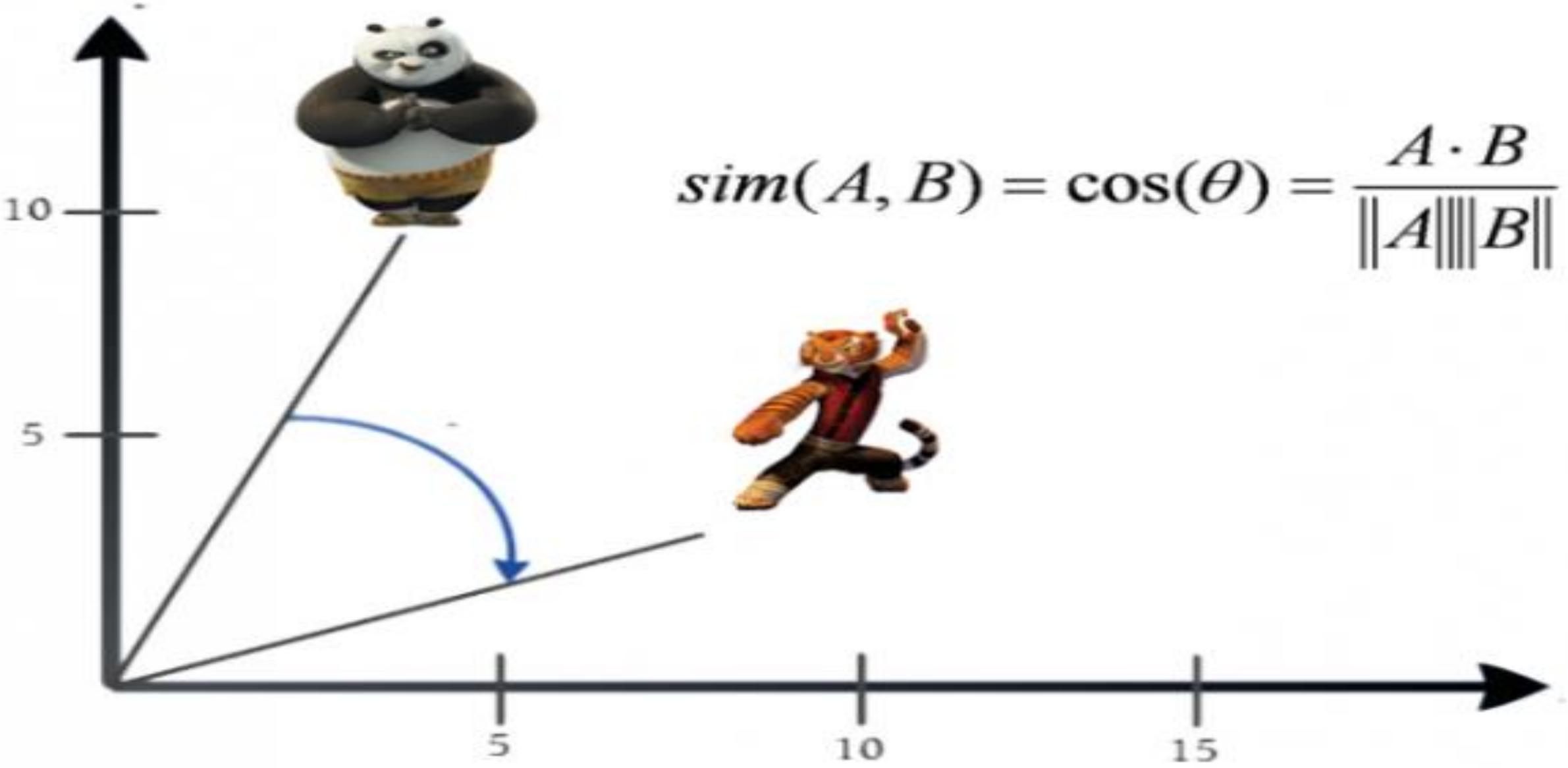
The algorithm generates recommendations based on few customers who are most similar to the user.

Here the similarity between the two customers is calculated using the cosine similarity.

The component vector are positive for purchased or positively rated items and negative for negatively rated items.

$$similarity(\vec{A}, \vec{B}) = \cos(\vec{A}, \vec{B}) = \frac{\vec{A} \bullet \vec{B}}{\|\vec{A}\| * \|\vec{B}\|}$$

Cosine Similarity



$$sim(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\|\|B\|}$$

LIMITATIONS

- Using collaborative filtering is computationally expensive
- Worst case complexity is O(MN)[M- Number of customers, N- Number. of product catalog].
- The final performance of the algorithm is O(M+N).
- The algorithm encounters severe performance and scaling issues.
- We can reduce M and N by a large factor but with the cost of recommendation quality.

Divide the customer base into many segments and assign the user to the segment containing the most similar customers

Segments are created using clustering.

The algorithm's goal is to assign the user to the segment containing the most similar customers. It then uses the purchases and ratings of the customers in the segment to generate recommendations.

Cluster models have better online scalability and performance than collaborative filtering because they compare the user to a controlled number of segments rather than the entire customer base.

CLUSTER MODELS

LIMITATIONS

- Cluster models group numerous customers together in a segment, match a user to a segment, and then consider all customers in the segment similar customers for the purpose of making recommendations.
- Because the similar customers that the cluster models find are not the most similar customers, the recommendations they produce are less relevant.
- online user–segment classification becomes almost as expensive as finding similar customers using collaborative filtering.

SEARCH-BASED METHODS

- Search- or content-based methods treat the recommendations problem as a search for related items.
- The algorithm constructs a search query to find other popular items by the same author, artist, or director, or with similar keywords or subjects.
- If the user has few purchases or ratings, search based recommendation algorithms scale and perform well.

GUY PEARCE CARRIE-ANNE MOSS JOE PANTOLIANO
MEMENTO

INSOMNIA

BATMAN BEGINS

THE PRESTIGE

THE DARK KNIGHT

INCEPTION

THE LEGEND ENDS
THE DARK KNIGHT RISES

INTERSTELLAR

LIMITATIONS

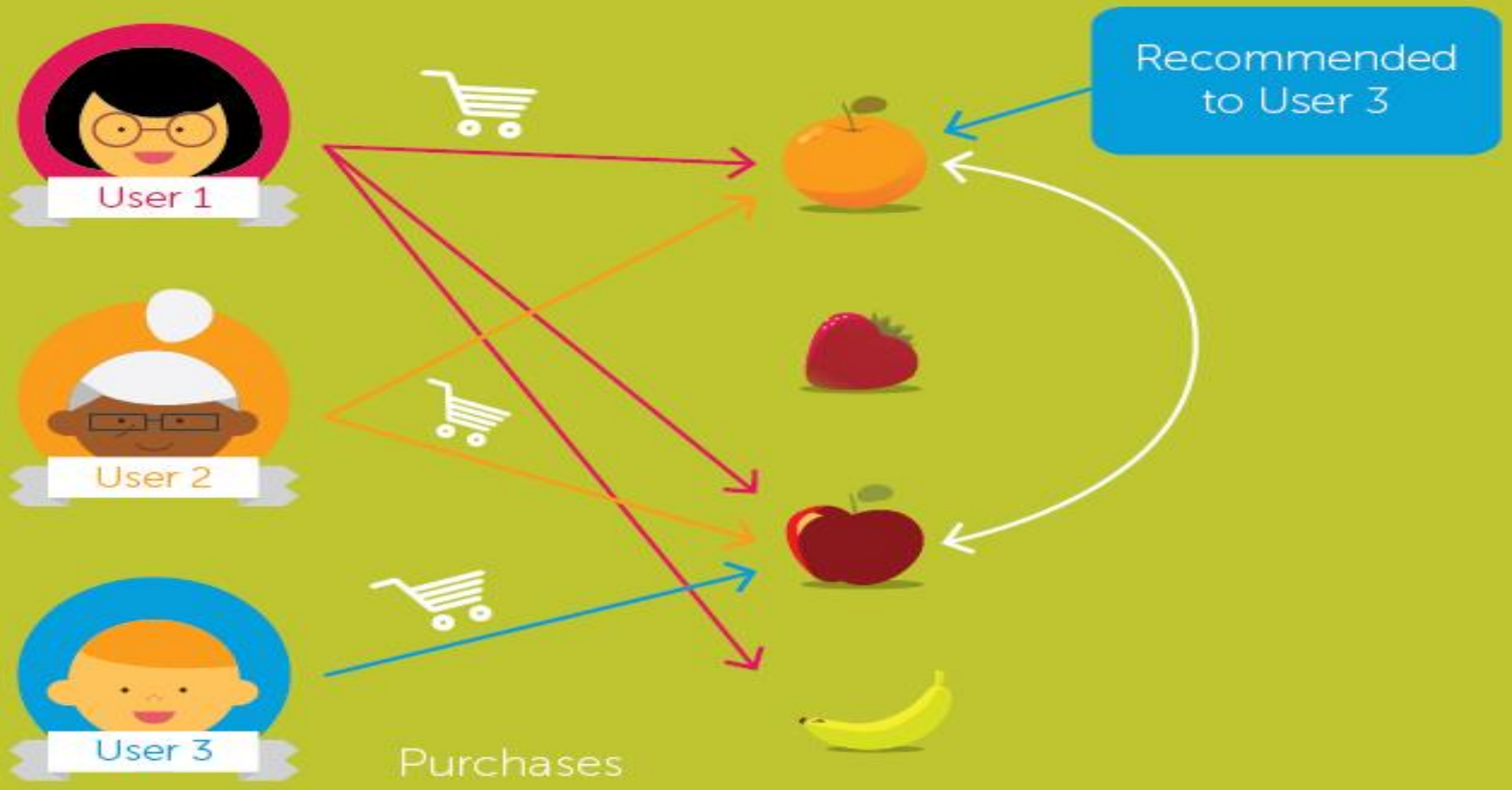
- For users with thousands of purchases, however, it's impractical to base a query on all the items.
- The algorithm must use a subset or summary of the data, reducing quality.
- In all cases, recommendation quality is relatively poor.
- Recommendations should help a customer find and discover new, relevant, and interesting items.

ITEM-TO-ITEM COLLABORATIVE FILTERING

- Rather than matching the user to similar customers, item-to-item collaborative filtering matches each of the user's purchased and rated items to similar items, then combines those similar items into a recommendation list.
- To determine the most-similar match for a given item, the algorithm builds a similar-items table by finding items that customers tend to purchase together.
- We could build a product-to-product matrix by iterating through all item pairs and computing a similarity metric for each pair. However, many product pairs have no common customers, and thus the approach is inefficient in terms of processing time and memory usage.

```
For each item in product catalog, I1
  For each customer C who purchased I1
    For each item I2 purchased by
      customer C
      Record that a customer purchased I1
      and I2
  For each item I2
    Compute the similarity between I1 and I2
```

Item-based filtering



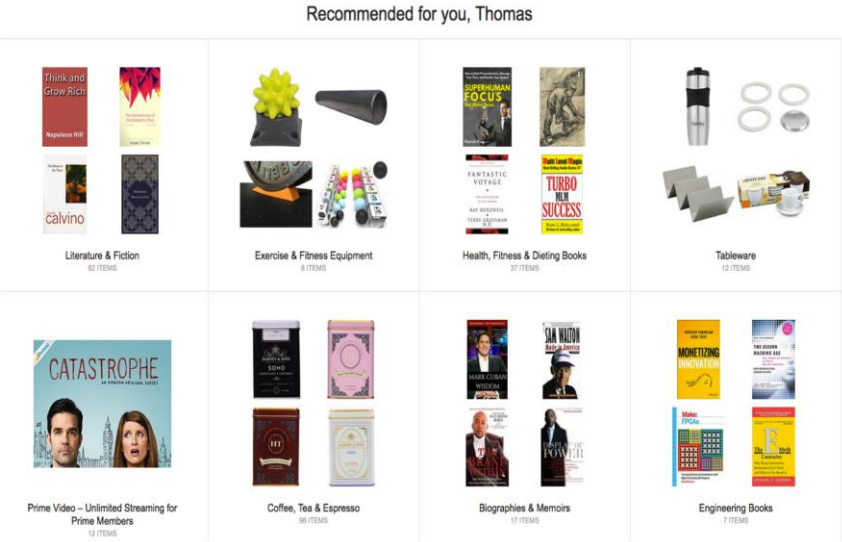
HOW IT WORKS

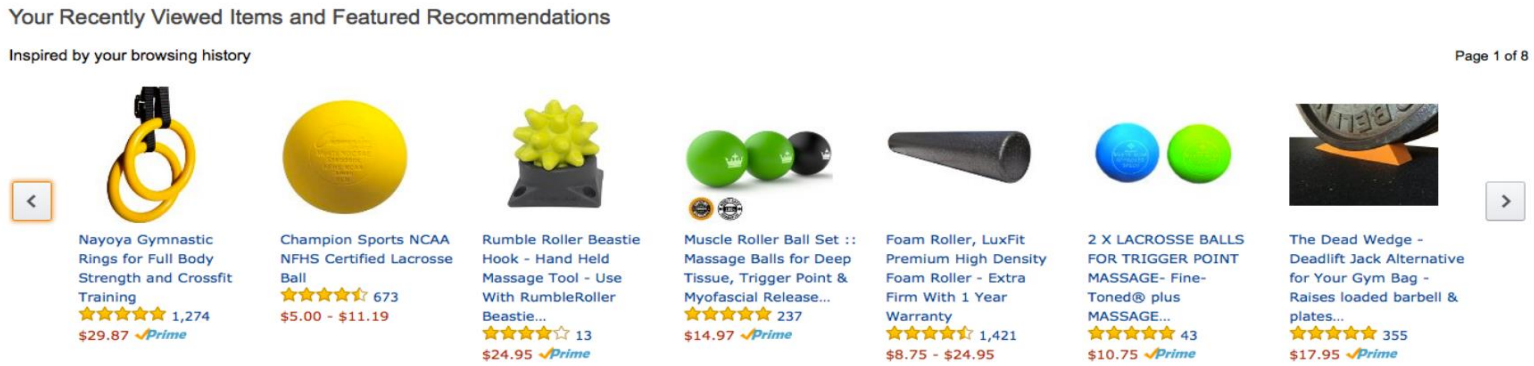
- It's possible to compute the similarity between two items in various ways, but a common method is to use the cosine measure, in which each vector corresponds to an item rather than a customer, and the vector's M dimensions correspond to customers who have purchased that item.
- This offline computation of the similar-items table is extremely time intensive, with O(N2 M) as worst case, but in real time it is O(NM).
- Given a similar-items table, the algorithm finds items similar to each of the user's purchases and ratings, aggregates those items, and then recommends the most popular or correlated items. This computation is very quick, depending only on the number of items the user purchased or rated.

SCALABILITY

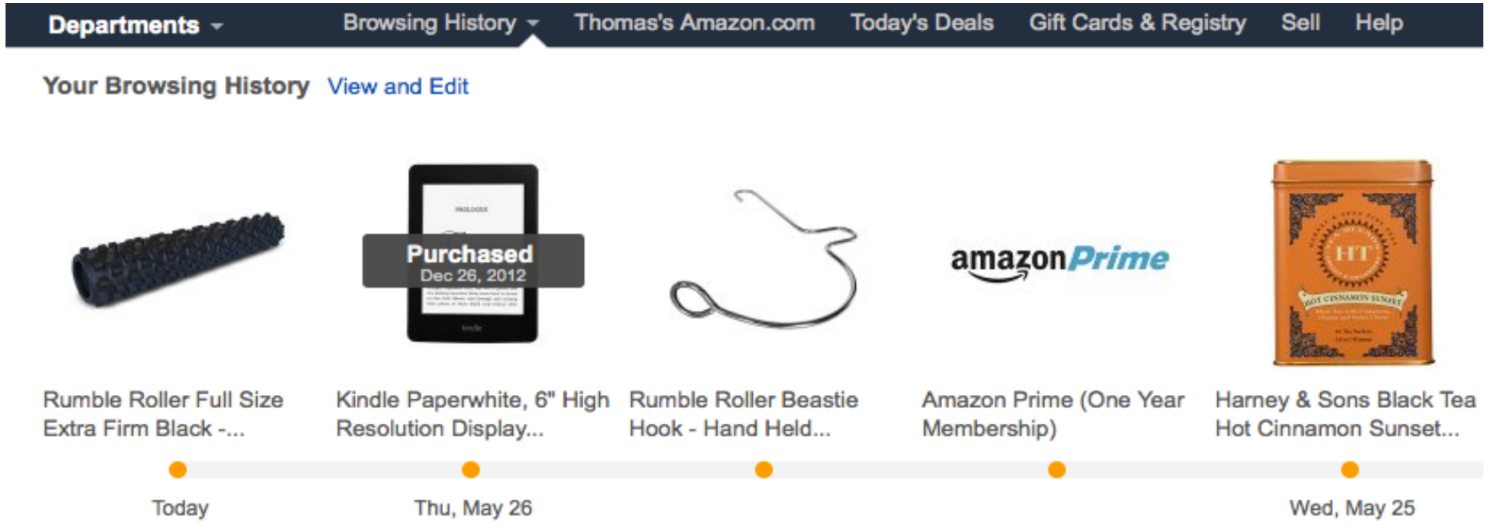
- Traditional collaborative filtering does little or no offline computation, and its online computation scales with the number of customers and catalog items.
- Cluster models can perform much of the computation offline, but recommendation quality is relatively poor. To improve it, it's possible to increase the number of segments, but this makes the online user–segment classification expensive.
- Search-based models build keyword, category, and author indexes offline, but fail to provide recommendations with interesting, targeted titles. They also scale poorly for customers with numerous purchases and ratings.
- For item-to-item it is dependent only on how many items the user has purchased or rated. Thus, the algorithm is fast even for extremely large data sets. Because the algorithm recommends highly correlated similar items, recommendation quality is excellent.

1) "RECOMMENDED FOR YOU, THOMAS"





Clicking on the "Your Recommendations" link on Amazon.com leads users to a page full of products recommended just for you. Amazon recommends a range of products from different categories you've been browsing, with the aim of putting products in front of you that you're likely to click, learn more about and buy.



Frequently Bought Together

Total price: $94.90

Add both to Cart

Add both to List

- This item: Rumble Roller - Textured Muscle Foam Roller Manipulates Soft Tissue Like A Massage Therapist $69.95
- Rumble Roller X-Firm Beastie and Base - Extra Firm Spiky Massage Ball - Comes With Base For... $24.95 ($3.12 / oz)

Best-selling emerging technology See more

There is a newer version of this item:

Kindle Paperwhite E-reader, 6" High-Resolution Display (300 ppi) with Built-in Light, Wi-Fi - Includes Special Offers

$119.99

(19,163)

In Stock.

Recommended for You Based on Kindle Paperwhite, 6" High Resolution Display w...

Page 1 of 5

MoKo Case for Kindle Paperwhite, Premium Thinnest and Lightest Leather Cover with...

898

$9.99 Prime

Swees Ultra Slim Leather Case Cover for Amazon All-New Kindle Paperwhite (Both 2012...

273

$3.99 Prime

Fintie SmartShell Case for Kindle Paperwhite - The Thinnest and Lightest Leather Cover for...

7,015

$14.99 Prime

Kindle Paperwhite, 6" High Resolution Display (212 ppi) with Built-in Light, Free 3G...

45,265

$159.99 Prime

Recommendation algorithms provide an effective form of targeted marketing by creating a personalized shopping experience for each customer.

Unlike other algorithms, item-to-item collaborative filtering is able to meet the challenges.

In the future, we expect the retail industry to more broadly apply recommendation algorithms for targeted marketing, both online and offline.

CONCLUSION

THANK YOU